

Samyuktha Donthiboina
Senior Java Full Stack Developer

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SUMMARY

- 7 years of experience as a Senior Full-Stack Software Developer specializing in **Java (8/11/17), Spring Boot, Spring (MVC, WebFlux, Cloud), Node.js, React.js, Angular 16, and microservices architecture.**
- Expertise in designing scalable, high-performance, cloud-native applications leveraging **AWS (EKS, ECS, EC2, S3, RDS, DynamoDB, Lambda, API Gateway, CloudFormation, CloudWatch, SageMaker) and GCP (Cloud Functions, Firestore).**
- Proficient in **RESTful & SOAP APIs, GraphQL**, and event-driven architectures using **Kafka, RabbitMQ, and ActiveMQ.**
- Hands-on experience in AI/ML frameworks: **TensorFlow, PyTorch, Keras, LangChain, Scikit-learn, MXNet, OpenAI, OpenCV, API integration, building LLM-powered applications**, retrieval-augmented generation (RAG), and automated document processing workflows.
- Strong knowledge of **multi-threading, concurrent programming, and parallel processing**, optimizing high-throughput applications.
- Experienced in **Oracle Coherence consulting, middleware integration, and distributed cache management**, ensuring high availability, cluster monitoring, and optimized application performance.
- Proficient in **Scala and Shell/Bash scripting** for automation, batch processing, and analytics workflows.
- Skilled in scripting and programming with **Python, Scala, Shell/Bash, JavaScript, C++, and C#.**
- Hands-on expertise in containerization and orchestration: **Docker, Kubernetes (EKS), Helm, Argo CD**, ensuring efficient deployments, **auto-scaling, and reliability.**
- Experienced in **CI/CD pipelines, Infrastructure as Code (IaC), and DevOps automation** with **Jenkins, GitHub Actions, AWS CodePipeline/CodeBuild/CodeDeploy, Terraform, and Argo CD.**
- Deep experience in **SQL & NoSQL databases** including **MySQL, PostgreSQL, MongoDB, Cassandra, Redis, Elasticsearch, and vector databases**, with performance tuning and query optimization expertise.
- Frontend expertise with **React.js, Angular 16, TypeScript, Redux, Material-UI**, building dynamic, responsive, and user-friendly interfaces.
- Strong understanding of **SDLC, Agile methodologies (Scrum, Kanban, Scrum-Ban)**, leading agile teams and DevOps transformation initiatives.
- Experienced in **maintaining and modernizing legacy applications using Struts and Log4J.**
- Experienced in secure application development implementing **OAuth2, JWT, Spring Security, API Gateway policies**, and circuit breaker patterns.
- Proven ability to modernize legacy monolithic systems into **microservices, improving scalability, maintainability, and cloud readiness.**
- Passionate about performance optimization, distributed systems, AI/ML integration, and delivering end-to-end enterprise solutions.
- Designed and implemented event-driven, asynchronous workflows using **Kafka, RabbitMQ, and ActiveMQ** for real-time notifications, high-throughput data processing, and **AI/ML pipeline orchestration.**
- Built and deployed **secure, scalable, cloud-native applications** with automated **CI/CD pipelines**, achieving significant reductions in deployment time and human intervention.
- Expertise in integrating **AI/ML models (TensorFlow, PyTorch, Keras)** with enterprise applications for **predictive analytics, anomaly detection, recommendation systems, and fraud detection.**
- Developed and optimized **GraphQL schemas and REST APIs** to improve frontend-backend data transfer efficiency, reducing latency and API overfetching.
- Extensive experience in **modernizing legacy monolithic applications** into microservices, including secure inter-service communication and multi-threaded, concurrent backend processing.
- Skilled in automated testing strategies including unit, integration, and functional testing with **JUnit, Mockito, Cypress, and Cucumber to ensure high-quality, production-ready code.**
- Led and mentored cross-functional teams, conducting code reviews, implementing best practices, and driving Agile ceremonies for successful software delivery.

- Hands-on experience in orchestrating containerized applications using **Docker, Kubernetes (EKS), Helm, and Argo CD for high availability, fault tolerance, and zero-downtime deployments.**
- Implemented serverless and cloud-native solutions using **AWS Lambda, API Gateway, DynamoDB, S3, and GCP Cloud Functions to optimize cost and scalability.**
- Developed **AI-powered document** processing, customer support, and knowledge retrieval systems leveraging **LangChain, RAG workflows, and OpenAI API integration.**

TECHNICAL SKILLS

- **Languages:** Java (8/11/17), J2EE (Servlets, JSP, Java Beans), Python, TypeScript, JavaScript, C#, C++, PL/SQL
- **Methodologies:** SDLC, Agile, Scrum-Ban, Kanban
- **Frameworks/Libraries:** Spring Boot, Spring (MVC, IOC, AOP, Cloud, WebFlux), Hibernate (4.x, 5.x), Struts (2.3/2.5), Log4J, JUnit, React.js, Angular 16, Redux, Material-UI
- **Web Technologies:** HTML, CSS, AJAX, Bootstrap, jQuery, XML, JSON, Node.js
- **AI/ML Frameworks:** TensorFlow, PyTorch, Keras, Scikit-learn, OpenCV, OpenAI API, LangChain
- **Web Services & APIs:** RESTful APIs, SOAP, GraphQL, Microservices
- **Event-Driven & Messaging:** Kafka, RabbitMQ, ActiveMQ
- **Databases:** MySQL, PostgreSQL, MongoDB, Cassandra, Redis, DynamoDB, vector databases
- **Middleware & Caching:** Oracle Coherence, WebLogic, JBoss
- **Cloud & DevOps:** AWS (EKS, ECS, EC2, S3, RDS, Lambda, API Gateway, CloudFormation, CloudWatch), GCP (GKE, Cloud Functions, Firestore), Docker, Kubernetes, Helm, Argo CD, Terraform, Jenkins, GitHub Actions, AWS CodePipeline/CodeBuild/CodeDeploy
- **Testing Tools:** JUnit, Mockito, Cypress, Selenium, Postman, Cucumber
- **Monitoring & Logging:** ELK Stack (Elasticsearch, Logstash, Kibana), Prometheus, Grafana, Coherence Cluster Console, JMX
- **Security:** OAuth2, JWT, RBAC, Spring Security

WORK EXPERIENCE

Client: Huntington Bank - Columbus, OH

Role: Senior Java Full Stack Developer

Aug 2024 – Present

Worked on **financial services applications**, focusing on **backend microservices** using Java, Spring Boot, and Scala for high-throughput transaction processing, AI-driven fraud detection, and batch analytics pipelines. Developed **frontend dashboards** in React.js and Angular integrating AI/ML insights for real-time reporting. Built **cloud-native solutions** on AWS with Terraform, Kubernetes, and serverless components, ensuring secure, scalable, and highly available systems.

Responsibilities:

- Developed and optimized **Java microservices using Spring Boot**, ensuring high scalability, fault tolerance, and robust service-to-service communication via REST APIs, GraphQL, and RabbitMQ.
- Integrated **Large Language Models (LLMs)** like GPT and BERT using LangChain for **AI-powered document processing**, automated customer support, and sentiment analysis workflows.
- Streamlined AI orchestration with LangChain for **retrieval-augmented generation (RAG)**, automated coding assistance, and knowledge retrieval pipelines.
- Designed and implemented **secure authentication and authorization** using Spring Security, OAuth2, and JWT for sensitive financial applications.
- Built **Python-based services** for real-time data ingestion, processing live financial data, integrating them into Java dashboards.
- Architected scalable **cloud infrastructure using Terraform**, automating provisioning of AWS resources including EC2, RDS, S3, Lambda, and VPCs for full-stack applications.
- Deployed microservices on **Kubernetes (EKS) with Helm charts**, implementing auto-scaling, rolling updates, and efficient resource management.
- Developed data processing and microservice components using **Scala** for high-performance batch processing pipelines.
- Maintained **legacy Struts-based modules** and integrated **Log4J** logging for monitoring, debugging, and traceability

in financial applications.

- Automated routine server and deployment tasks using **Shell/Bash scripts**, reducing manual intervention and improving efficiency.
- Optimized backend performance using **multi-threading, concurrency, JDBC**, and asynchronous processing techniques to handle high-throughput requests efficiently.
- Implemented distributed **Cassandra** database solutions for low-latency, high-throughput storage of transactional and analytics data.
- Designed and optimized distributed caching solutions using **Oracle Coherence**, improving transaction throughput and reducing database load.
- Implemented **middleware integration** between legacy financial applications and modern microservices using WebLogic and Spring Boot.
- Configured and monitored **Coherence clusters**, ensuring high availability, failover resilience, and efficient in-memory data grid performance.
- Automated **cluster monitoring and alerting** using Coherence JMX metrics integrated with Prometheus and Grafana dashboards.
- Optimized backend performance using **multi-threading, concurrency, and asynchronous processing** techniques to handle high-throughput requests efficiently.
- Developed **AI-driven predictive analytics, recommendation systems, and anomaly detection models** using TensorFlow, PyTorch, and Keras.
- Enhanced observability and monitoring using **ELK Stack, Prometheus, and Grafana**, setting up dashboards, alerts, and logs for improved performance tracking.
- Used **Postman** extensively to test **REST/GraphQL APIs** and validate microservice integrations during development and QA cycles.
- Integrated **C# legacy services with Java microservices** for eligibility verification, insurance claims processing, and data aggregation pipelines.
- Integrated **OpenCV with Python services** to preprocess document images and extract structured data, supporting AI-driven fraud detection and automated document verification workflows.
- Designed and optimized **GraphQL schemas and queries** to improve frontend data retrieval performance by 40% and reduce API overfetching.
- Implemented **unit, integration, and functional testing** using JUnit, Mockito, and Cucumber, ensuring robust code coverage and release quality.
- Built **event-driven architectures** with Kafka, RabbitMQ, and ActiveMQ to enable asynchronous workflows, real-time notifications, and messaging reliability.
- Automated CI/CD pipelines leveraging **AWS CodePipeline, CodeBuild, and CodeDeploy**, reducing deployment time and human intervention.
- Implemented **API Gateway policies**, rate-limiting, circuit breaker patterns, and authentication to ensure secure and fault-tolerant microservices.
- Developed **AI-based fraud detection and risk assessment models** to reduce false positives and enhance operational efficiency.
- Optimized **SQL and NoSQL databases** (PostgreSQL, MongoDB, Redis, DynamoDB, vector databases) for low-latency retrieval and high-performance data processing.
- Collaborated with frontend teams to develop **dynamic React.js and Angular interfaces**, integrating AI/ML insights and real-time analytics dashboards.
- Mentored junior developers, conducted code reviews, and implemented best practices for **secure, maintainable, and scalable codebases**.
- Designed **asynchronous batch processing modules** for large-scale data ingestion, processing, and AI model inference.
- Enhanced **containerized application deployments** using Docker, Kubernetes, Helm, and Argo CD, ensuring high availability, fault tolerance, and zero-downtime deployments.
- Conducted **performance tuning** across backend services, AI pipelines, and database layers, improving throughput and reducing latency.
- Ensured **documentation and knowledge sharing** for AI/ML workflows, microservices, and cloud infrastructure to streamline team collaboration and onboarding.

Environment/Technologies Used: Java 17, Spring Boot, Spring Cloud, React.js, Angular 16, Python, LangChain, Kafka, RabbitMQ, ActiveMQ, JDBC, AWS (EKS, EC2, Lambda, RDS, S3), Terraform, Docker, Kubernetes, Helm, Argo CD, TensorFlow, PyTorch, Keras, PostgreSQL, MongoDB, Redis, DynamoDB, vector databases, ELK Stack, GitHub Actions, Scala, Shell/Bash, Struts 2.x, Log4J, Cassandra, Postman

Client: SSM Health - Madison, WI

Role: Senior Java Developer

Aug 2023 - Jul 2024

Developed **healthcare applications** handling real-time patient and clinical data. Implemented **backend microservices** in Java and Python for predictive analytics, AI chatbots, and fraud detection, while maintaining HIPAA compliance. Built **frontend components** using React.js and Angular with dynamic dashboards. Automated **CI/CD pipelines** and managed **cloud infrastructure** on AWS/GCP for high availability and scalability.

Responsibilities:

- Developed high-performance **Java 17 microservices with Spring Boot** for healthcare applications, processing large volumes of real-time patient and clinical data.
- Built **RESTful and GraphQL APIs** for AI-powered applications, integrating predictive analytics, patient risk scoring, **fraud detection in insurance claims, and chatbot services using Python and TensorFlow.**
- Migrated monolithic legacy healthcare services to microservices architecture on **AWS EKS and GCP GKE**, improving resilience, scalability, and deployment efficiency.
- Implemented event-driven architectures with **Kafka, RabbitMQ, and AWS SQS** for real-time monitoring, anomaly detection, and patient data processing.
- Developed AI-assisted chatbots for healthcare patient support, reducing average handling time by 40% and improving user satisfaction.
- Automated deployments using **GitHub Actions, Jenkins, and Terraform, enforcing IaC standards** for repeatable and secure infrastructure provisioning.
- Designed secure file transfer and document management systems using **AES/RSA** encryption and Bouncy Castle libraries, ensuring **HIPAA compliance.**
- Developed **Scala-based** services for healthcare analytics pipelines, improving performance in patient data processing tasks.
- Managed serverless cloud-native healthcare applications using **AWS Lambda, API Gateway, DynamoDB, and S3** for highly scalable and cost-efficient solutions.
- Created Python-based ML models for **predictive analytics, fraud detection, and risk assessment**, integrated into Java microservices via REST APIs.
- Integrated **middleware messaging and data-sharing solutions** across healthcare systems, leveraging Oracle Coherence and WebLogic for reliable data synchronization.
- Monitored and fine-tuned **Coherence distributed caches** for real-time patient and claims data processing, achieving sub-millisecond response times.
- Ensured cluster stability by implementing **load-balancing, failover, and partition recovery strategies** within Coherence grids.
- Applied **OpenCV for medical document image preprocessing** and anomaly detection pipelines, improving accuracy of ML models for patient risk scoring and claims validation.
- Optimized frontend performance using **React.js, Angular 16, TypeScript, Redux, and Material-UI**, implementing lazy loading and code-splitting techniques.
- Developed multi-threaded, concurrent modules for secure batch processing, patient transaction validation, and data-intensive healthcare operations.
- Created automation scripts using **Shell/Bash** to schedule batch jobs, backups, and cloud deployment tasks.
- Maintained **CI/CD pipelines** with automated test execution and shift-left testing strategies, improving code quality and reducing production defects.
- Designed AI-powered search and retrieval systems using vector embeddings, improving query accuracy and response times for patient records.
- Maintained older **Struts modules** and enhanced application logging with **Log4J** for effective issue tracking and auditing.
- Improved database performance with **PostgreSQL, MongoDB, MySQL, Redis, implementing indexing, partitioning, and caching strategies.**

- Implemented **OAuth2/JWT authentication and RBAC** for secure **API access** and compliance with HIPAA and healthcare data security standards.
- Developed event-driven monitoring and alerting systems for AI pipelines and real-time healthcare dashboards.
- Migrated legacy **ETL processes** to modern microservices architecture, improving maintainability and processing efficiency for clinical and operational data.
- Built serverless Lambda functions for background processing, notifications, and automated reporting workflows.
- Developed Dockerized microservices and orchestrated deployments with Kubernetes, ensuring high availability and fault tolerance.
- Designed and maintained **Cassandra keyspaces** for storing patient and clinical event data with high availability.
- Configured **ELK Stack and Splunk** for centralized logging, performance monitoring, and AI-based anomaly detection.
- Collaborated with cross-functional teams to integrate **AI/ML** insights into enterprise healthcare applications for enhanced decision-making.
- Validated **RESTful and GraphQL** endpoints with **Postman**, ensuring robust API functionality and contract compliance.
- Mentored junior developers on **microservices, cloud deployment, and AI/ML integration**, improving overall team productivity.
- Conducted **performance tuning, profiling, and optimization** for backend services, AI workloads, and database queries.
- Maintained end-to-end documentation and knowledge base for applications, AI pipelines, and cloud infrastructure automation.

Environment/Technologies Used: Java 17, Spring Boot, Node.js, Python, React.js, Angular 16, TensorFlow, PyTorch, Kafka, RabbitMQ, AWS (EKS, Lambda, DynamoDB, API Gateway, S3, SQS), GCP (GKE, Cloud Run), Terraform, Docker, Kubernetes, Jenkins, GitHub Actions, PostgreSQL, MongoDB, Redis, Scala, Shell/Bash, Struts 2.x, Log4J, Cassandra, Postman

Client: SITA – Bohemia, NY

Role: Full Stack Developer

Jan 2022 – Jul 2023

Worked on Telecommunication solutions, developing **backend microservices in Java, Spring Boot, and gRPC** for data and booking management. Built **frontend interfaces** using **React.js, TypeScript, and Redux** for dynamic, responsive user experiences. Implemented event-driven architecture with **Kafka/RabbitMQ**, and deployed applications on AWS using Kubernetes and serverless Lambda functions.

Responsibilities:

- Participated in **Agile** development cycles, collaborating with cross-functional teams and contributing to sprint planning, stand-ups, and retrospective meetings.
- Architected and developed full-stack applications using **Java 11, Spring Boot, Spring MVC, Spring WebFlux, Spring Cloud, React.js, Redux**, ensuring modular and maintainable codebases.
- Implemented microservices architecture, designing scalable and loosely coupled services using **Spring Cloud, Spring WebFlux, and gRPC** to enhance performance and inter-service communication.
- Developed GraphQL APIs for optimized data querying, improving API flexibility and reducing over-fetching of data.
- Designed and optimized RESTful APIs using **Spring Boot, JPA, and Hibernate**, improving data retrieval efficiency and integration with front-end applications.
- Integrated **AWS DynamoDB and RDS** for handling high-volume transactional data, leveraging GSI and LSI indexing for efficient query performance.
- Developed scalable event-driven architectures using Apache Kafka and RabbitMQ, ensuring reliable asynchronous processing of messages.
- Managed asynchronous operations with **RxJS Observables and Operators**, improving application responsiveness and handling concurrent user requests effectively.
- Designed and implemented **OAuth 2.0 and OpenID Connect (OIDC) authentication** for secure access management across microservices.
- Developed and maintained CI/CD pipelines using Jenkins, Docker, and Kubernetes, enabling automated testing and seamless deployment across environments.

- Leveraged **Terraform and AWS CloudFormation** to automate infrastructure provisioning, ensuring repeatable and consistent cloud deployments.
- Utilized **Redis caching for reducing API** response times and improving application throughput.
- Enhanced system observability and logging using ELK Stack (Elasticsearch, Logstash, Kibana) **and AWS CloudWatch**, ensuring proactive monitoring and alerting for production systems.
- Implemented **AWS Lambda** functions for serverless computing, automating background tasks and reducing infrastructure costs.
- Deployed containerized applications using Kubernetes (EKS), managing deployments, scaling, and service discovery efficiently.
- Worked with AWS API Gateway to expose microservices securely, handling rate limiting and authorization policies.
- Implemented service discovery and load balancing using Spring Cloud Netflix Eureka and Ribbon, optimizing traffic distribution.
- Integrated **Swagger and OpenAPI** documentation, ensuring consistent and well-documented API specifications for internal and external teams.
- Designed and developed front-end components using **React, TypeScript, Redux Toolkit, and Material-UI**, improving code modularity and maintainability.
- Utilized Webpack and Babel for optimizing front-end bundling and improving performance in React applications.
- Developed complex forms and UI components with **React Hooks and Context API**, ensuring stateful behavior and component reusability.
- Implemented lazy loading and code splitting in React applications, enhancing page load performance and optimizing resource utilization.
- Migrated monolithic applications to microservices-based architectures, reducing technical debt and improving scalability.
- Implemented real-time data synchronization using **Web Socket, Socket.IO**, improving interactivity in collaborative applications.
- Optimized SQL and NoSQL databases (**MySQL, PostgreSQL, MongoDB**) by implementing indexing strategies, partitioning, and query optimization techniques for improved data access performance.
- Built role-based access control (RBAC) and permission management for securing user interactions within the application.
- Designed and implemented unit and integration tests using **JUnit, Mockito, Jest, and Cypress**, ensuring application reliability and minimizing regressions.
- Conducted code reviews, mentoring junior developers, and enforcing best practices in software development and system design.
- Managed cloud security by configuring **AWS IAM** roles, policies, and security groups, ensuring compliance with security best practices.

Environment/Technologies Used: Agile, Spring Boot, Spring MVC, Spring WebFlux, Spring Cloud, React.js, Redux, Java 11, JPA, Hibernate, Terraform, AWS CloudFormation, AWS IAM, OAuth 2.0 and OpenID Connect (OIDC), JDBC, Redis, JUnit, Mockito, Jest, Cypress, Web Socket, Socket.IO

Client: Hinduja Global Solution - Bangalore, India

Role: Software Engineer

May 2018 – Oct 2021

Worked on health insurance applications for **Aetna**, building **backend microservices** using Java, Spring Boot, and microservices for **claims processing, patient eligibility, and reporting workflows**. Developed **frontend modules** using React.js and Redux for interactive client portals. Implemented **event-driven and batch processing solutions** with Kafka and Spring Batch, and automated deployments with Docker, Kubernetes, and CI/CD pipelines.

Responsibilities:

- Designed and implemented **microservices architecture** using Spring Boot, ensuring modularity, scalability, and maintainability.
- Developed **backend services** using Java, Spring Boot, Spring Security, Spring Data JPA, RESTful APIs, SOAP, and GraphQL for optimized data retrieval and secure integrations.
- Implemented **JWT-based authentication** and **role-based access control (RBAC)**; developed OAuth 2.0 flows for secure third-party integrations.

- Built **event-driven architectures** using Kafka and AWS SQS for asynchronous message processing and system scalability.
- Developed **batch processing applications** using Spring Batch for large-scale data ingestion and processing.
- Optimized **database performance** with PL/SQL, Hibernate second-level caching (EHCache), indexing, and partitioning strategies.
- Developed **message-driven applications** using RabbitMQ and Kafka, ensuring reliable asynchronous processing.
- Integrated third-party APIs for **payment processing, external authentication, and real-time data synchronization**.
- Managed **legacy SOAP and RESTful integrations**, ensuring interoperability with older systems.
- Developed **interactive front-end modules** using React.js, Redux, and Context API for client portals.
- Created **reusable React components** (functional & class-based) with state management, lazy loading, and code splitting for improved performance.
- Implemented **real-time data synchronization** using WebSockets and Server-Sent Events (SSE).
- Integrated **Swagger/OpenAPI documentation** for consistent API references with frontend teams.
- Managed **containerized applications** with Docker and Kubernetes for high availability and scalability.
- Automated infrastructure provisioning using **CI/CD pipelines (Azure DevOps, Jenkins)** and **serverless deployments** with AWS Lambda and Azure Functions.
- Utilized **API Gateway** and Load Balancers for traffic distribution and system availability.
- Followed **Test-Driven Development (TDD)** with JUnit and Mockito; performed **end-to-end UI testing** using Selenium and Cypress.
- Conducted **code reviews, refactoring, and performance tuning** for optimized backend and frontend performance.
- Applied **caching strategies** using Redis to reduce response time and improve system performance.
- Led Agile practices including **sprint planning, backlog grooming, and daily stand-ups** to ensure efficient project execution.
- Maintained **version control** with Git/GitHub for efficient collaboration and change tracking.

Environment/Technologies Used: Java, Spring Boot, Spring MVC, Spring Security, Spring Data JPA, React.js, Redux, RESTful APIs, SOAP, Docker, Kubernetes, AWS Lambda, Azure Functions, JDBC, Kafka, AWS SQS, Git, GitHub, JUnit, Mockito, Cypress, PL/SQL, EHCache, WebSockets, SSE, OAuth 2.0, Spring Batch